

Abstract

Streaming Data Analysis and Visualization

Streaming Data Analysis and Visualization focuses on collecting, processing, analyzing, and displaying data continuously as it is generated from sources such as Internet of Things (IoT) devices, social media platforms, financial transactions, sensors, and web applications.

This project focuses on **household energy consumption data** by simulating a continuous data stream and extracting meaningful insights from appliance energy readings. The system uses **Python, Pandas, and NumPy** for data cleaning, statistical analysis, streaming simulation, running and moving averages, anomaly detection, and peak consumption identification. **Matplotlib** is used to create time-based and statistical visualizations such as line charts, bar charts, histograms, and box plots.

The project loads the household energy consumption CSV dataset, combines the Date and Time fields into a DateTime value, converts appliance readings into numeric values, removes incomplete and duplicate records, and sorts the data chronologically. Statistical measures such as mean, median, minimum, maximum, and standard deviation are calculated. The first 100 appliance readings are processed as a simulated data stream, and a moving average is calculated to observe consumption trends. Potential anomalies are identified using a **mean $\pm$ 2 $\times$ standard deviation** threshold. Peak consumption and hourly consumption patterns are also analyzed.

The results demonstrate how data analysis and visualization can help identify **energy consumption trends, peak usage periods, and unusual readings**. Overall, the project provides a basic streaming-style analytical pipeline for understanding household energy usage and supporting better energy management and decision-making.

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Google Colab
- CSV Dataset

Key Concepts

Streaming Data Analysis · Household Energy Consumption · Data Cleaning · Statistical Analysis · Moving Average · Anomaly Detection · Peak Consumption · Data Visualization · Time-Series Analysis